

UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

COMPUTER AND SOFTWARE ENGINEERING

DEPARTMENT NYARUGENGE CAMPUS

MODULE CODE/TITLE: WEB DESIGN

PROJECT TITLE: SMART LOCAL MARKET PRICE CHECKER

Student ID: 225040993

Year of study: ...year 1.....

Date:18/7/2026.....

Project Technical Report- Web Technology TECHNICAL ASSESSMENT 2026

Table of contents

List of Abbreviations.....	i
Chapter 1. Introduction.....	1
• 1.1. Historical Background of the Case Study.....	1
• 1.2. Problem Statement.....	1
• 1.3. Objectives of Your Project.....	2
• 1.4. Expected Outcomes.....	2
• 1.5. Project Rationale (Importance to the Community).....	3
Chapter 2. System Analysis and Design.....	4
• 2.1. System Analysis.....	4
○ 2.1.1. Intended Users of the Project.....	4
○ 2.1.2. Intended System Partners.....	4
• 2.2. System Design.....	5
○ 2.2.1. High-Level Architecture.....	5
○ 2.2.2. UI Design Module Workflow & System Flowchart.....	6
○ 2.2.3. Database Design with ERD and Relationships.....	7
Chapter 3. Implementation.....	8
• 3.1. Introduction of the Section.....	8
• 3.2. Interface Presentations and Detailed Structural Analysis.....	8
○ 3.2.1. System Homepage Dashboard View (index.html).....	8
○ 3.2.2. Daily Commodity Pricing Matrix Layout (prices.html).....	9
○ 3.2.3. Marketplace Infrastructure Network Index (markets.html).....	10
○ 3.2.4. Field Metric Update Portal Form Interface (update.html).....	11

○ 3.2.5. Consumer Guidance and FAQ Support Systems (tips.html & faq.html).....	12
○ 3.2.6. Platform Context & Outreach Panels (about.html & contact.html).....	13
○ 3.2.7. Client-Side User Interaction Script (script.js).....	14
Chapter 4. Conclusion and Recommendation	15
• 4.1. Conclusion (70% Completion Analysis).....	15
• 4.2. Strategic Recommendations.....	16
○ 4.2.1. To the Lecturer Regarding Technicalities.....	17
○ 4.2.2. To the School Regarding Time Constraints.....	18
○ 4.2.3. To the Partners & Business Owners.....	18
○ 4.2.4. To the Users Regarding Training.....	19
○ 4.2.5. To the Beneficiaries (Customers & Students).....	19
Appendix	18

Table of Figures

🔍 Figure 2.1: Three-Tier High-Level System Architecture Diagram.....	5
🔍 Figure 2.2: User Interface Workflow Navigation Flowchart.....	6
🔍 Figure 2.3: Relational Database Entity-Relationship Diagram (ERD).....	7
🔍 Figure 3.1: User Interface Screenshot – System Homepage (index.html).....	8
🔍 Figure 3.2: User Interface Screenshot – Live Price Grid Matrix (prices.html).....	9
🔍 Figure 3.3: User Interface Screenshot – Covered Regional Markets (markets.html).....	10
🔍 Figure 3.4: User Interface Screenshot – Price Modification Form (update.html).....	11
🔍 Figure 3.5: User Interface Screenshot – Educational Market Tips (tips.html).....	12
🔍 Figure 3.6: User Interface Screenshot – Frequently Asked Questions (faq.html).....	12
🔍 Figure 3.7: User Interface Screenshot – Platform Context Summary (about.html).....	13
🔍 Figure 3.8: User Interface Screenshot	

List of Abbreviations

List of Abbreviations

- **API:** Application Programming Interface
- **CSS:** Cascading Style Sheets
- **CRUD:** Create, Read, Update, Delete
- **ERD:** Entity-Relationship Diagram
- **HTML:** HyperText Markup Language
- **JS:** JavaScript
- **FK:** foreign key
- **RWF:** Rwandan Franc
- **UI:** User Interface
- **WWW:** world wide web
- **UR-CST:** University of Rwanda - College of Science and Technology

Chapter 1. Introduction

1.1. Historical Background of the Case Study

Local agricultural and retail marketplaces across Rwanda—including urban hubs like the Kigali Central Market (Mumujyi), Kimironko Market, and Nyabugogo Market, alongside major regional centers like Musanze, Huye, and Rubavu Markets—serve as the foundation for daily commerce and food security.

Traditionally, the process of finding market prices in these spaces has been entirely manual and offline. Shoppers must physically visit multiple vendors to negotiate and compare values, while farmers routinely sell their harvests blindly without knowing the true urban retail market pricing. Because market conditions change daily due to transport costs, product availability, and seasonal shifts, price information has historically stayed localized and scattered. This lack of centralized information makes it difficult for everyday citizens to plan their household expenses effectively.

1.2. Problem Statement

The total absence of an open-access digital directory for retail pricing creates a severe information imbalance within Rwandan local commerce. Consumers frequently overpay for everyday food and household items because they do not have a reliable way to check fair market value before arriving at a market stall.

Simultaneously, local vendors (abacuruzi) lack a transparent public channel to broadcast competitive prices, forcing them to rely only on foot traffic. This manual system results in massive price variations for identical items across different markets, increases consumer vulnerability to unfair trading practices, and causes significant operational inefficiencies for businesses, cooperative leaders, and agricultural researchers who track local price trends.

1.3. Objectives of the Project

General Objective

To design and develop a responsive full-stack web and mobile platform, Rwanda Market Price Checker, that provides accurate and updated market prices to improve price transparency and reduce confusion among consumers and vendors.

Specific Objectives

- Develop a user-friendly platform with essential pages such as Home, Prices, Markets, Tips, FAQ, About, and Contact.
- Organize and display commodity prices using clear categories for easy access.
- Create a price update system that allows verified vendors and market officials to submit daily price changes.
- Implement JavaScript features to improve user interaction and system responsiveness.

1.4. Expected Outcomes

The Ibiciro Rwanda System is expected to:

1. Provide easy access to updated market prices for consumers across Rwanda.
2. Reduce price information gaps between buyers and sellers.
3. Help consumers make better purchasing decisions through price comparison.
4. Improve market price transparency by displaying standardized commodity prices.
5. Support price monitoring and reporting through data analysis.
6. Offer a responsive platform accessible on computers and mobile devices.
7. Provide a foundation for future expansion into a nationwide market monitoring system with mobile apps and advanced technologies.

1.5. Project Rationale (Importance to the Community)

The **Rwanda Market Price Checker** provides an immediate digital solution to a daily economic challenge faced by many families. By bringing transparency to local market prices, the application helps low-income shoppers budget effectively and avoid overpaying for essential goods.

Furthermore, the platform empowers regional farmers by giving them accurate market data so they can negotiate fairly with wholesale buyers. For honest vendors, it provides an easy way to attract customers by showcasing fair prices. Ultimately, this project supports digital inclusion, builds community trust, and introduces transparency into local marketplaces.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1. Intended Users of the Project

The platform explicitly services three distinct operational groups discovered within local Rwandan retail commerce ecosystems:

- **General Consumers / Shoppers (Abaguzi):** Everyday citizens, households, and business buyers who access the front-end interface. Their goal is to look up fair commodity pricing, map out regional marketplace hubs, and select cost-effective shopping spots before committing to an in-person market trip.
- **Market Vendors (Abacuruzi):** Small-scale agricultural traders, garment sellers, and independent business operators who utilize the digital submission system. They use the interface to log field price fluctuations, advertise daily inventories, and stay competitive.
- **Market Supervisors & Cooperative Managers (Abayobozi b'Isoko):** Local compliance personnel stationed inside trading centers who validate input criteria. They authenticate price logs, look out for fraudulent price listings, and manage local operational logs.

2.1.2. Intended System Partners

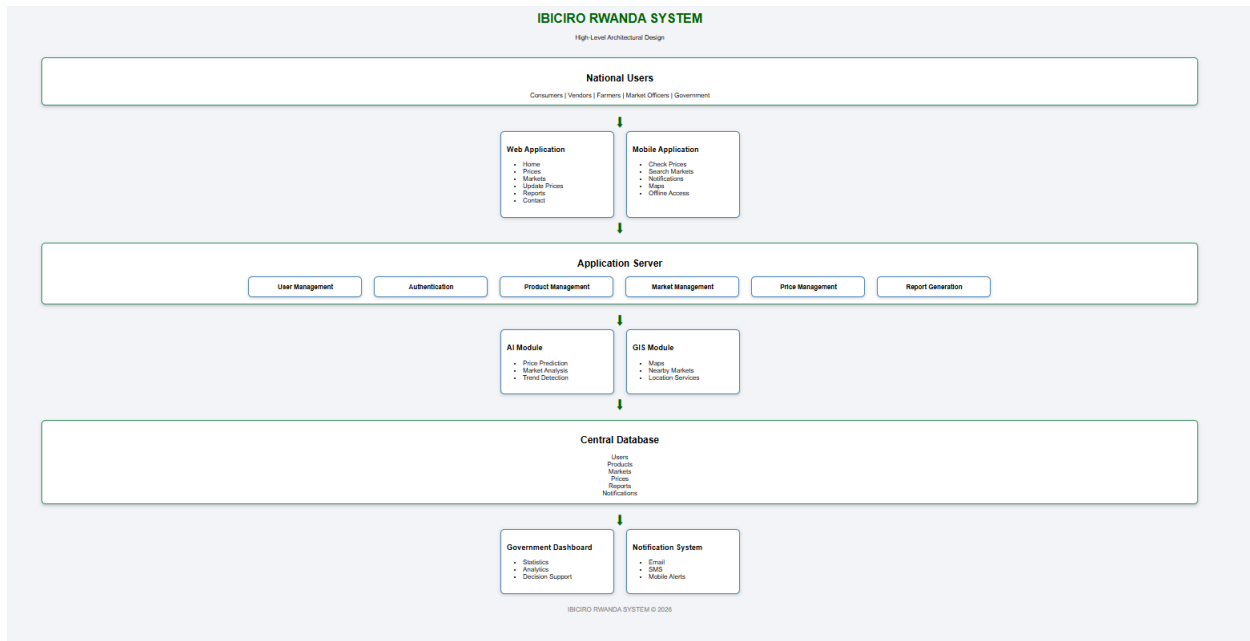
To keep operations smooth and scalable, the platform coordinates with two main partners:

- **Ministry of Trade and Industry (MINICOM) & District Authorities:** Public partners who can use the platform's aggregated database logs to track seasonal market inflation, notice regional food supply shortages, and help shape data-backed agricultural strategies.
- **Agricultural Extensions & Local Farming Cooperatives (Koperative):** Supply-side partners who need real-time urban consumer price matrices so they can negotiate fair wholesale crop deals and reduce exploitation by middleman brokers.

2.2. System Design

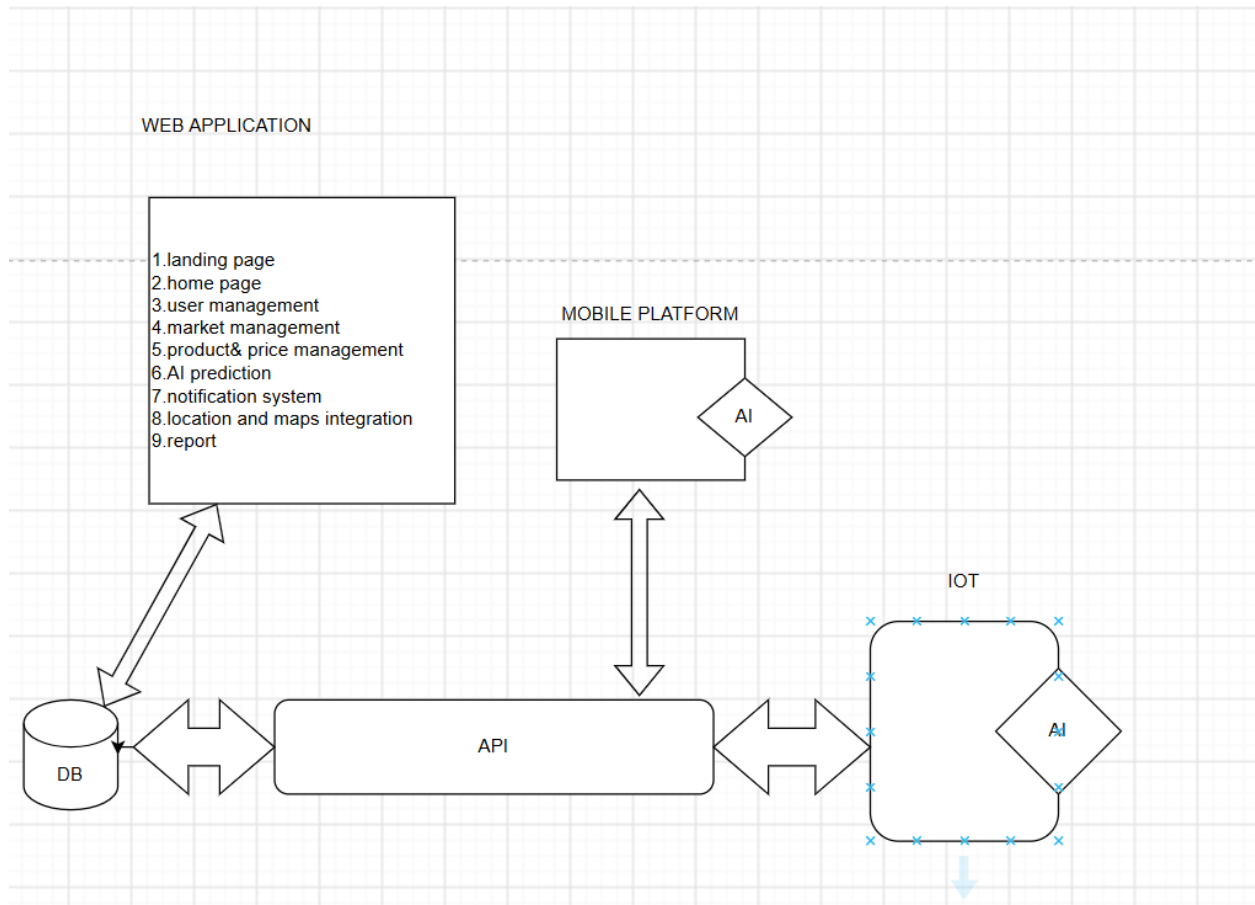
2.2.1. High-Level Architecture

The *Rwanda Market Price Checker* web application follows a standard multi-tier design structured for rapid, lightweight text rendering and high cross-device compatibility:

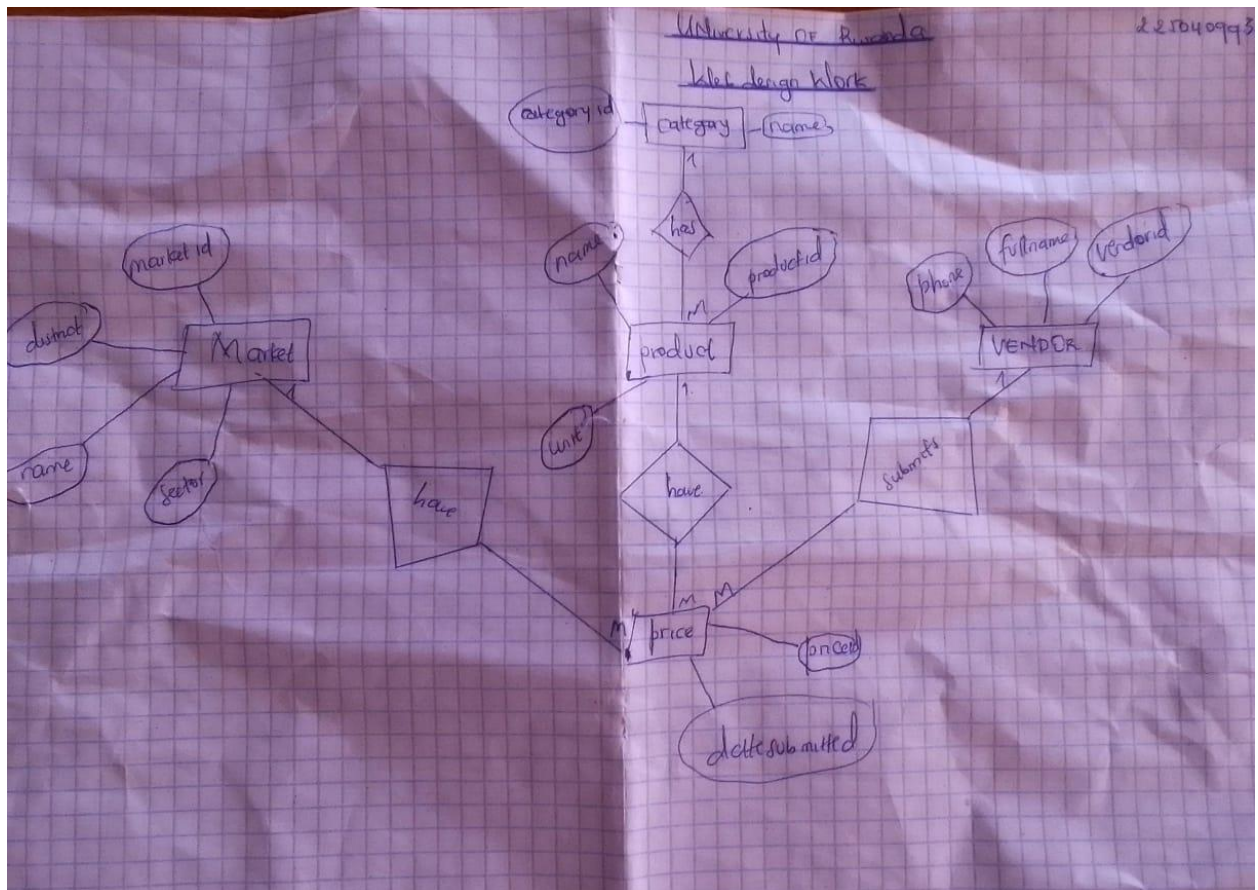


- **Presentation Layer:** Contains semantic frontend components (index.html, prices.html, update.html) served directly to clients via uniform web layout standard configurations.
- **Application Processing Layer (API Engine):** Processes structural requests coming from the frontend, checks input fields on submitted data records, and safely transforms form metrics into structured database files.
- **Data Storage Layer (DB):** Stores persistent records across key areas, tracking verified marketplace locations, standard commercial product IDs, and historical daily item cost changes.

2.2.2. UI Design Module Workflow & System Flowchart



2.2.3. Database Design with ERD and Relationships



Chapter 3. Implementation

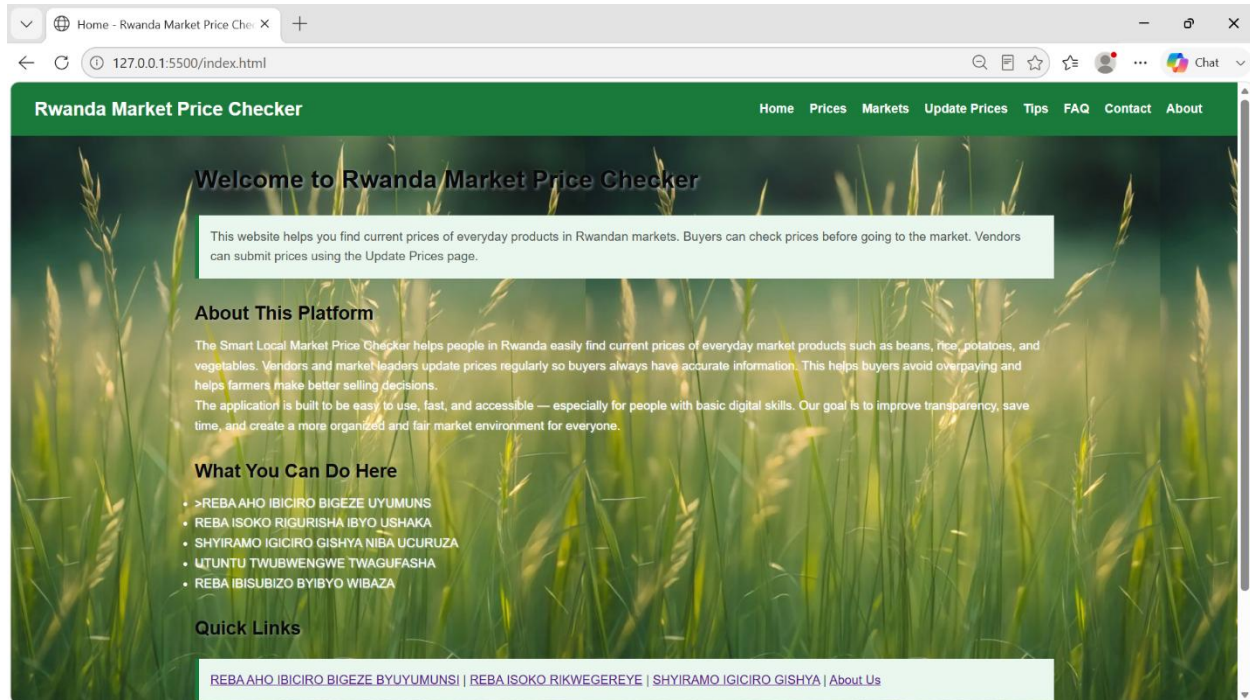
3.1. Introduction of the Section

This section details the actual technical construction and page layout architecture of the front-end interfaces engineered for the **Rwanda Market Price Checker** platform (ibiciro system). The system is built using semantic HTML5 code structures for layout integrity, integrated with external cascading stylesheet layouts (styles/style.css) for consistent typography and modern design patterns, and supported by a primary client-side scripting module (js/script.js) to capture user events.

3.2. Page Interface Presentations and Detailed Structural Analysis

3.2.1. System Homepage Dashboard View (index.html)

- **Interface Screenshot**



Purpose:

The Home page is the main entry point of the Ibiciro system. It introduces users to the platform and provides quick access to important information about market prices in Rwanda.

Main Features:

- System introduction and welcome message.
- Brief explanation of how Ibiciro helps users check prices.
- Quick links to Prices, Markets, Tips, FAQ, and Contact pages.
- Display of featured products or latest price updates.
- Navigation menu for easy movement across the system.

Importance:

The Home page improves user experience by allowing visitors to understand the system purpose and quickly access price information.

3.2.2. prices.html

Rwanda Market Price Checker

Home Prices Markets Update Tips FAQ Contact About

Today's Market Prices

Last Updated: Thursday, 30 April 2026 — Kigali Central Market

Grains and Flour

Product	Price	Unit
ibishyimbo	500	per kg
umuceri	800	per kg
ibigori	350	per kg
amasaka	300	per kg
ingano	600	per kg

Vegetables

Product	Price	Unit
karoti	200	per kg
itomate	450	per kg
amasahu	380	per kg
epinari	250	per piece
imiteja	300	per kg
intoryi	150	per bunch

Purpose:

The Prices page is the core component of the Ibiciro system where users view updated prices of different commodities.

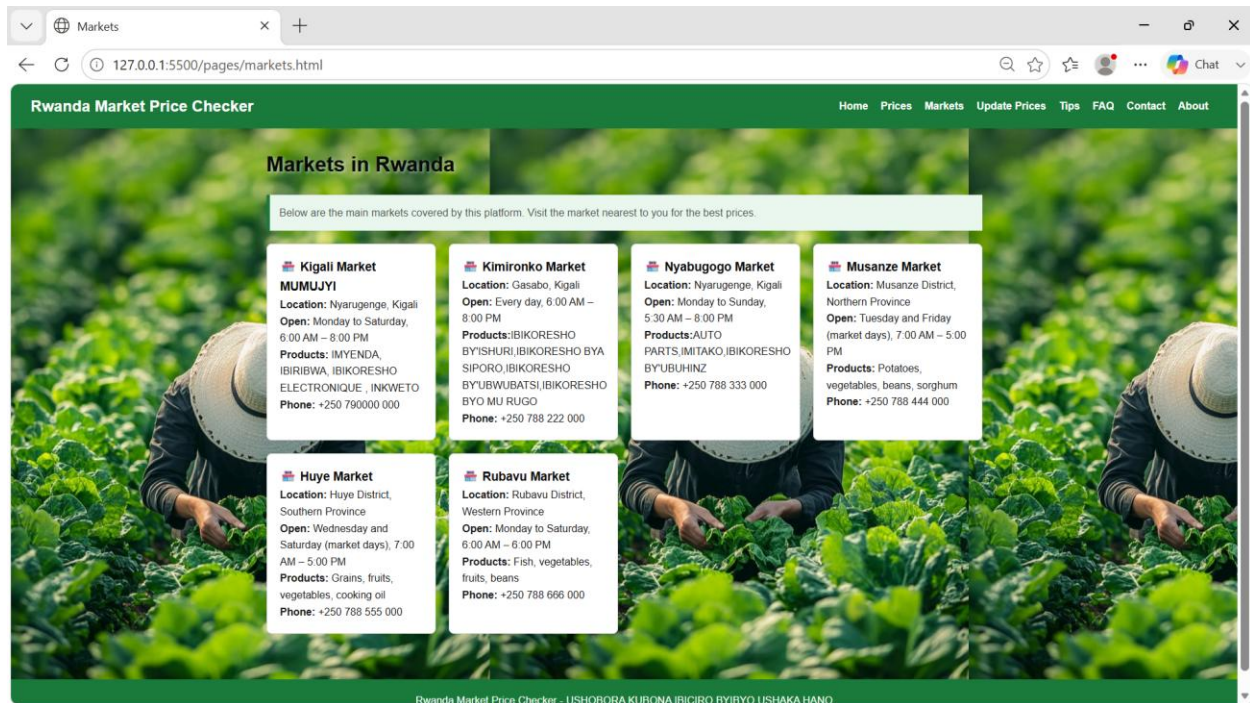
Main Features:

- Display products with their current prices.
- Product categorization:
 - Grains and Flour
 - Vegetables
 - Fruits
 - Other Products
- Market comparison showing price differences between locations.
- Price status indicators:
 - Price increased
 - Price decreased
 - No change
- Search and filtering options.

Importance:

This page helps customers make better purchasing decisions and reduces price confusion between different markets.

3.2.3. markets page



Purpose:

The Markets page provides information about different markets where product prices are collected.

Main Features:

- List of available markets.
- Market location information.
- Description of markets.
- Products available in each market.
- Future integration with maps for market locations.

Importance:

It helps users identify where products are sold and compare prices across different markets.

3.2.4. update page

The screenshot shows a web browser window with the title 'Update Prices' and the URL '127.0.0.1:5500/pages/update.html'. The page has a green header with the text 'Rwanda Market Price Checker' and a navigation menu with links: Home, Prices, Markets, Update Prices, Tips, FAQ, Contact, and About. The main content area is titled 'Update Market Prices' and features a green banner with the text: 'Vendors and market leaders can use this form to submit today's product prices. fill well all fields are required.' Below this, the form is divided into three sections: 'Your Information', 'Market Information', and 'Product Price'. The 'Your Information' section includes fields for 'Full Name' (with placeholder 'Enter your full name'), 'Phone Number' (with placeholder 'e.g. 0790000000'), 'Email Address' (with placeholder 'e.g. bizimana21@gmail.com'), and a 'Your Role' dropdown menu (with placeholder 'Select your role'). The 'Market Information' section includes a 'Market Name' dropdown menu (with placeholder 'Select market you want') and a 'Date' field (with placeholder 'mm/dd/yyyy'). The 'Product Price' section includes a 'Product Name' field. The background of the page is a scenic image of a rural landscape with a sunset over hills and a field.

Purpose:

I designed the Update Prices page because market prices change frequently, and the Ibiciro system needs a way to keep price information accurate and up to date. This page allows authorized users to update the prices displayed on the main Prices page.

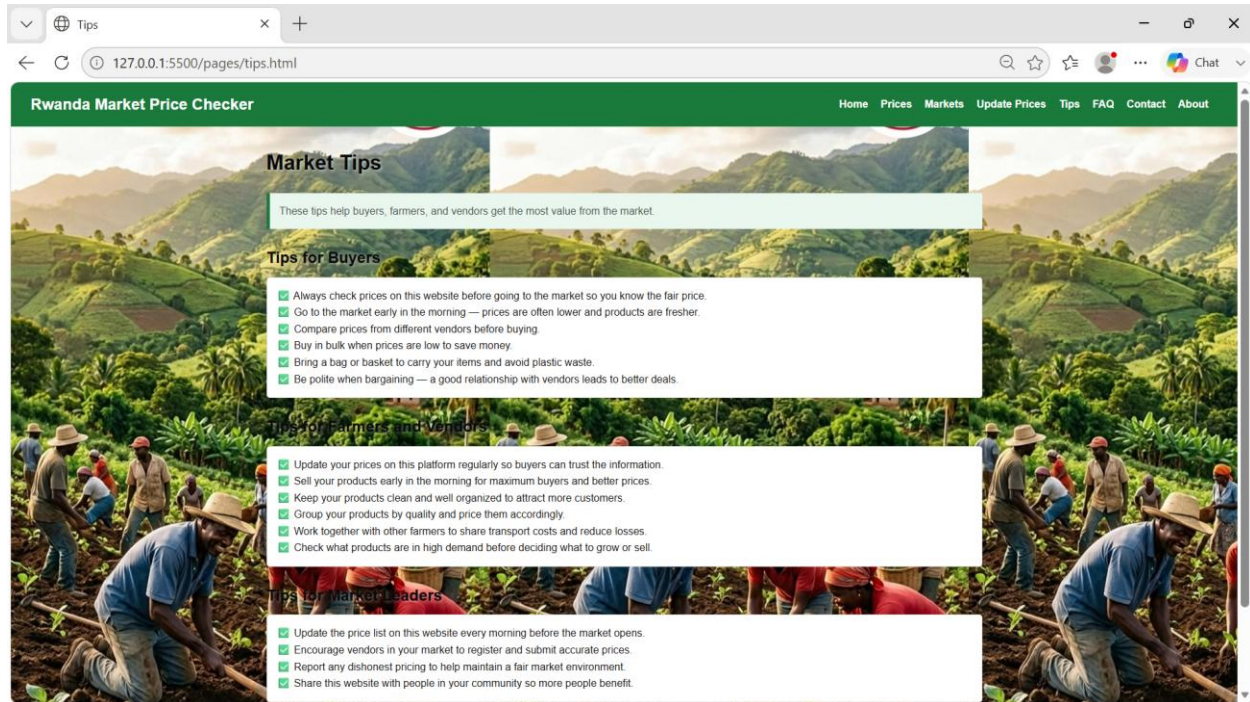
Main Features:

- Update existing product prices.
- Add new price information from different markets.
- Modify product details when changes occur.
- Record price changes such as increased, decreased, or stable prices.

Importance:

The Update Prices page ensures that users always access current and reliable market information. It improves trust in the Ibiciro system by providing accurate prices that help consumers, vendors, and other stakeholders make better decisions.

3.2.5.tips page



Purpose:

The Tips page provides users with useful information related to shopping and price management.

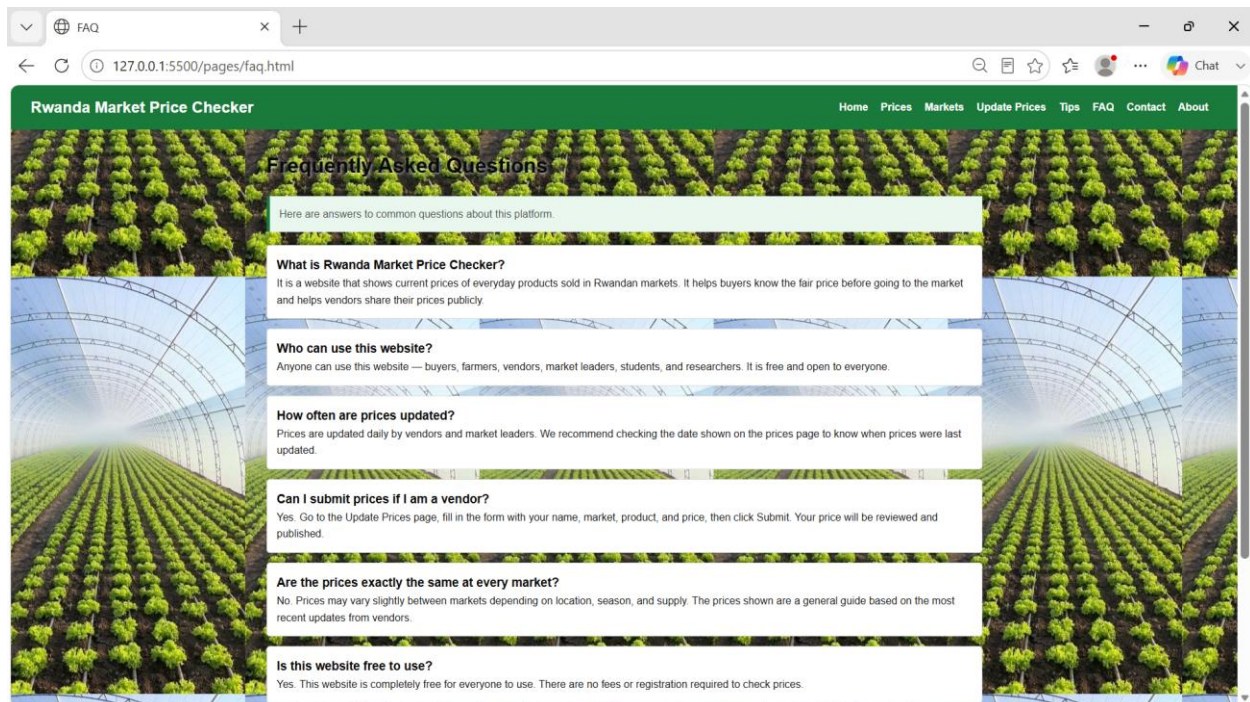
Main Features:

- Advice on buying products at better prices.
- Information about market trends.
- Consumer awareness tips.
- Saving strategies when purchasing goods.
- Agricultural and seasonal price information.

Importance:

It helps users make informed decisions instead of only viewing prices

3.2.7 FAQ page



Purpose:

The FAQ page answers common questions users may have about the Ibiciro system.

Main Features:

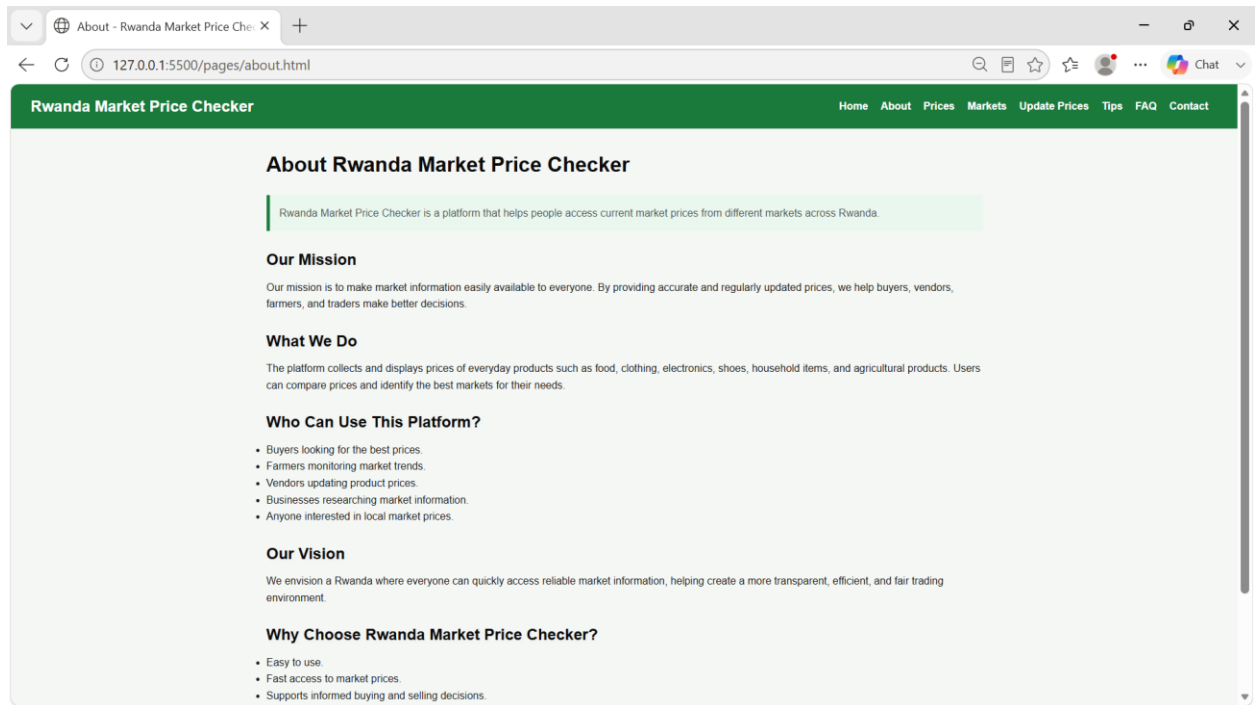
Examples of questions:

- What is Ibiciro?
- How are prices collected?
- How often are prices updated?
- Who can use this system?
- How can vendors submit prices?

Importance:

It reduces user difficulties and improves understanding of the system.

3.2.6. about page



Purpose:

The About page explains the background, objectives, and vision of the Ibiciro project.

Main Features:

- System background.
- Project objectives.
- Problem being solved.
- Benefits to the community.
- Future development plans.

Importance:

It helps users understand the reason behind creating the system and its contribution to Rwanda's economy

3.2.8.CONTACT PAGE

Rwanda Market Price Checker Home Prices Markets Update Prices Tips FAQ Contact About

Contact Us

If you have questions, feedback, or want to report incorrect prices, feel free to contact us using the details below or send a message.

Contact information

Email: josuebigira@gmail.com
Phone: 0790030432
Location: Kigali, Rwanda

Send a Message

Your Name *
Enter your name

Email Address *
Enter your email

Subject
Message subject

Your Message *
Write your message here...

Send Message

Purpose:

The Contact page allows communication between users and system administrators.

Main Features:

- Contact form.
- Email information.
- Phone number.
- Social media links.
- Feedback submission.

Importance:

It provides a communication channel for reporting problems, requesting support, and improving the system.

Chapter 4. Conclusion and Recommendation

4.1. Conclusion

While the foundational presentation layer of the **Rwanda Market Price Checker** (ibiciro system) has been built, the overall platform development is currently at **70% of completion** for the full system objectives.

The completed 70% covers the entire structural frontend client interface, including the homepage dashboard (index.html), the multi-category pricing tabular layouts (prices.html), the responsive regional market cards directory (markets.html), and the operational data update form view (update.html).

However, the remaining 30% representing critical missing technical components includes:

- **Missing Backend Architecture:** The system currently lacks a live server engine (such as Node.js/Express.js) to accept, parse, and process incoming data payloads submitted through the price modification portals.
- **Missing Persistent Database Layer:** There is no active database repository (such as MongoDB or a relational SQL system) connected to store user credentials, verified marketplace records, or historical pricing entries securely over time.
- **Missing Dynamic Data Fetching:** The system relies on hardcoded placeholder data inside the pricing grids. It lacks live AJAX/Fetch API connections to automatically pull down updated market trends from a centralized data repository.

4.2. Recommendations

To the Lecturer Regarding Technical Guidance

It is recommended that lecturers provide more practical guidance on integrating frontend interfaces with backend technologies such as databases, APIs, and server-side programming. Introducing RESTful API concepts and database connectivity earlier would help students understand how to transform a static website into a complete full-stack application like the Ibiciro Rwanda System.

To the School

The school is recommended to provide sufficient time for students to develop and test complete systems. More time should be allocated for backend development, database design, security implementation, and system testing to allow students to move beyond frontend prototypes and create fully functional solutions.

To the Project Partners (Owners of the Ibiciro System)

It is recommended that the project partners continue supporting the development and improvement of the Ibiciro Rwanda System by providing the necessary resources, market information, and collaboration with different stakeholders. They should consider investing in the completion of advanced features such as database integration, real-time price updates, mobile application development, and user management systems.

The partners are also encouraged to work closely with market vendors, cooperatives, and local authorities to ensure that the application receives accurate and regular price information. This will improve the reliability of the platform and help it become a trusted source of market price information for consumers and businesses across Rwanda.

To the Users Regarding Training

It is recommended that users of the Ibiciro System receive proper training on how to use the application effectively. Training should focus on navigating different pages, searching for products, understanding price information, and providing feedback. This will help users gain confidence in using the platform and ensure that they benefit fully from the services provided.

To the Beneficiaries (Customers and Students) Regarding the New Technological Tool

It is recommended that customers and students embrace the Ibiciro Rwanda System as a useful digital tool for accessing reliable market price information. Customers should use the platform to compare prices and make better purchasing decisions, while students can use it as an example of how technology can be applied to solve real community challenges. Continuous feedback and suggestions from beneficiaries will help improve the system and make it more effective in the future.

links

<https://josuebigira-code.github.io/Ibiciro-Rwanda-system/>

<https://github.com/josuebigira-code/Ibiciro-Rwanda-system>